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COURSE: ITA0609 - MACHINE LEARNING

CO3 ASSESSMENT TOOL 1- Experiments

1.Naïve Bayes Classifier

Aim:

To implement a Naïve Bayes classifier, evaluate its performance using accuracy, precision, and recall, and compare the results with and without Laplace smoothing.

Algorithm:

1. Import the dataset.
2. Split the data into training and testing sets.
3. Train the Naïve Bayes classifier.
4. Predict the test data.
5. Calculate Accuracy, Precision, and Recall.
6. Repeat using Laplace smoothing.
7. Compare the results.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score

x, y = load_iris(return_X_y=True)

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=1)

model = GaussianNB()
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, average="macro"))
print("Recall:", recall_score(y_test, y_pred, average="macro"))
```

Input:

Iris dataset

Output:

Accuracy: 0.9777

Precision: 0.978

Recall: 0.978

Result:

The Naïve Bayes classifier was implemented successfully and achieved high accuracy, precision, and recall.

2. Maximum Likelihood Estimation (MLE)

Aim:

To estimate the mean and variance of a dataset using Maximum Likelihood Estimation (MLE).

Algorithm:

1. Create a sample dataset.
2. Calculate the mean.
3. Calculate the variance.
4. Display the estimated parameters.

Code:

```
import numpy as np
```

```
data = [10, 12, 13, 15, 18]
```

```
mean = np.mean(data)
```

```
variance = np.var(data)
```

```
print("Mean =", mean)
```

```
print("Variance =", variance)
```

Input:

10 12 13 15 18

Output:

Mean = 13.6

Variance = 7.44

Result:

The parameters of the dataset were estimated successfully using the Maximum Likelihood Estimation (MLE) method. The estimated mean is 13.6 and the variance is 7.44.